

Omar Fathy ? Embedded Systems Engineer

Giza, Egypt | omar.fathy@email.com

SKILLS

Languages: C, C++, Assembly (ARM Cortex-M)

Platforms: STM32, Arduino, Raspberry Pi, FreeRTOS

Protocols: UART, SPI, I2C, CAN bus

Tools: Keil MDK, STM32CubeIDE, Oscilloscope, Logic Analyzer, Git

Practices: RTOS programming, interrupt handling, low-level hardware debugging

Soft Skills: analytical thinking, precision, hardware-software integration

EXPERIENCE

Embedded Engineer ? Valeo Egypt (2020?present)

- Developed AUTOSAR-compliant firmware for automotive ECUs in C
- Implemented CAN bus communication protocol for vehicle sensor data
- Debugged hardware issues using oscilloscope and logic analyser

Junior Embedded Developer ? ITI (2019?2020)

- Programmed STM32 microcontrollers for IoT sensor nodes
- Worked with FreeRTOS for task scheduling

EDUCATION

BSc Electronics and Communication Engineering ? Ain Shams University (2019)